EXERCISE SET 2.3

Use truth tables to determine whether the argument forms in 6–11 are valid. Indicate which columns represent the premises and which represent the conclusion, and include a sentence explaining how the truth table supports your answer. Your explanation should show that you understand what it means for a form of argument to be valid or invalid.

9.

$$
\begin{array}{l}
\quad p \wedge q \rightarrow \sim r \\
\quad p \vee \sim q \\
\quad \sim q \rightarrow p \\
\therefore \sim r
\end{array}
$$

p	**q**	**~q**	**p ∧ q**	**p ∧ q → ~r**	**p ∨ ~q**	**~q → p**	**~r**
T	T	F	T	F	T	T	F
T	T	F	T	T	T	T	T
T	F	T	F	T	T	T	F
T	F	T	F	T	T	T	T
F	T	F	F	T	F	T	F
F	T	F	F	T	F	T	T
F	F	T	F	T	T	F	F
F	F	T	F	T	T	F	T

$$
\left.\begin{array}{l}
\quad p \wedge q \rightarrow \sim r \\
\quad p \vee \sim q \\
\quad \sim q \rightarrow p
\end{array}\right\} \text{Premises}
$$

$$
\left.\begin{array}{l}
\therefore \sim r
\end{array}\right\} \text{Conclusion}
$$

I've marked the critical rows of the truth table, where all premises of the argument form are true.

In order for the argument form to be valid, the truth set for the conclusion has to align with the truth values of the same set of the premises.

We can therefore deduce from the truth table that the argument form is invalid, since the truth set of the conclusion includes a false value in one of the critical rows.

Solution: The argument form is invalid.